

# Nilancy Agarwal

✉ nilancyagarwal@gmail.com    ☎ 9257603310    📍 Chennai, Tamil Nadu    🔗 linkedin  
🐙 github    📁 portfolio    🏆 LeetCode



## EDUCATION

**SRM Institute Of Science and Technology**  
B.Tech in Computer Science, CGPA: 9.75/10

2023 – 2027 | Chennai, India

## SKILLS

**Programming Languages:** C++, Java, Python, JavaScript

**Computer Science Fundamentals:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN), Software Engineering

**Web Development:** React.js, Node.js, Express.js, REST APIs, HTML5, CSS3, Tailwind CSS

**Databases & Cloud:** MySQL, MongoDB, Firebase, AWS

**Tools & Version Control:** Git, GitHub, CI/CD

**AI & ML:** TensorFlow, PyTorch, CNNs, Vision Transformers (ViT), NLP, LLMs, RAG, Embeddings

## PROFESSIONAL EXPERIENCE

**Full Stack Developer Intern – Sikar Infotech**

05/2025 – 07/2025

**Tech Stack:** React.js, PHP, MySQL, REST APIs, Git

- Developed modular **full-stack applications** with secure authentication and transaction handling, improving system reliability and maintainability.
- Developed and enhanced **two production-level applications**, implementing RESTful APIs and payment gateway workflows.
- Optimized database schema design and backend queries, reducing **API response time by ~30–40%** and improving overall application responsiveness.

**AI/ML Intern – Infosys Springboard**

08/2025 – 10/2025

**Tech Stack:** Python, LLMs, RAG, NLP, Embeddings, Pandas

- Developed BudgetWise, an AI-driven financial analytics platform using **RAG based LLM pipelines** to process unstructured financial data and automate transaction categorization with **80–85%** classification accuracy.
- Built Python-based data preprocessing, automation, and embedding retrieval workflows, improving contextual relevance of **responses by 25%** and reducing manual data processing efforts.
- Implemented **NLP powered summarization** modules to automatically generate structured financial insights from raw transactional data, reducing manual analysis effort by over **50%**.

## RESEARCH EXPERIENCE

**Hybrid GANs and ViT Approach for Cloud Removal in Sentinel-2** 📄

12/2025 – 02/2026 | Chennai, India

(Accepted at IEEE International Conference on Innovations in Engineering and Technology (ICIET 2026))

- Developed a **hybrid deep learning framework** integrating **UNet, GAN, and Vision Transformer (ViT)** architectures for cloud removal in Sentinel-2 satellite imagery, enabling high-fidelity image reconstruction under dense cloud occlusions.
- Built an **automated geospatial data pipeline** for retrieval, preprocessing, and patch extraction, achieving **PSNR 33.3 dB** and **SSIM 0.96** across **250+** image patches.

**ECG Classification Using Multi-Block CNN** 📄

03/2025 – 05/2025 | Chennai, India

(Presented at the **Third International Conference on Networking and Communications**)

- Designed and trained a **four-block CNN architecture** for **ECG arrhythmia classification** on the **MIT-BIH dataset**, leveraging signal preprocessing and feature engineering for robust heartbeat analysis.
- Achieved **99.1% classification accuracy** with **45 ms inference latency** through model optimization, class-balancing strategies, and systematic hyperparameter tuning.

## PROJECTS

**DTC Trading Club -Stock Market Platform** 📄

**Tech Stack:** React.js, PHP, MySQL, Financial APIs, Razorpay

- Developed and deployed a **full-stack financial platform** enabling real-time stock visualization, portfolio tracking, and market analytics for users.
- Designed a **dynamic backend-driven CMS workflow** allowing administrators to manage trading data and website content without redeployment, improving operational efficiency.
- Integrated **secure payment processing (Razorpay)** with order validation and transaction logging, ensuring reliable membership registrations and seamless user experience.

**Digitalized Local Shopping** 📄

**Tech Stack:** React, Node.js, MySQL

- Developed a **location-based marketplace platform** enabling users to discover nearby stores and products within a configurable search radius.
- Implemented **product comparison, review, and rating systems** to support informed purchasing decisions and enhance user engagement.
- Built a **centralized product catalog and search functionality**, enabling efficient product discovery across multiple local vendors.

## **ACHIEVEMENTS & CERTIFICATIONS**

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- **Citi Bridge'26 (Runner-Up) at CitiBank** – Top 80/5000+ applicants; led team to Runner-Up position in a 24-hour AI Fintech Promptathon.
- Solved **300+** Data Structures and Algorithms problems on LeetCode.
- SRMJEEE **AIR 28** – Awarded **100% Merit** Scholarship on tuition fees.
- HackOdisha 5.0 – Selected among the **Top 5** teams and awarded a cash prize.
- NPTEL Software Testing (**Gold Elite, Top 1%**) | NPTEL Machine Learning (**Silver Elite, Top 5%**)
- AWS Certified Solutions Architect – Associate | AWS Certified AI Practitioner